WILSON

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Head of Technology Development — Payments, Digital Channels & Integration | Regulated Delivery · Hands-On Engineering

PROFESSIONAL SUMMARY

Technology leader with 18+ years of engineering and management experience who leads multi-discipline teams and delivers BAU and change projects to deadline, currently in a MAS-regulated Tier-1 bank. Began as a Java engineer and technical lead on consumer platforms at Renren, Sohu and Baidu, then spent 2016–2022 in head-of-technology and architect roles at start-ups and scale-ups.

Since 2022, a Vice President in UOB's Group Technology & Operations, leading 25 onshore and offshore engineers on two platform lines — cross-border payments and retail unsecured lending — that serve millions of accounts. Delivers change projects alongside BAU production support, governs IT service vendors and applies MAS TRM (Singapore technology-risk rules) through secure SDLC and data governance.

Brings customer-channel experience from consumer internet and start-ups: e-commerce sites and their payment-gateway integrations, a news CMS, a local-life platform with card and e-wallet payments, and a B2B SaaS platform behind an API gateway. Remains hands-on in Java — reviewing new core pull requests daily, co-designing solutions with engineers and writing core modules for significant changes.

CORE COMPETENCIES

Technology Leadership	Multi-discipline teams via squad leads and engineering managers; mentoring and performance management; onshore/offshore, cross-time-zone teams; 5+ years managing offshore teams
Delivery & Prioritisation	BAU and change projects in parallel, prioritisation across two platform lines, roadmaps, milestones and deadlines, upgrades and migrations, risk and issue tracking, change control, steering-committee reporting
Channels & Integration	E-commerce websites, news-portal CMS, API gateway, B2B and B2C platforms, channel and core-banking integration, payment gateways (Alipay, WeChat Pay, UnionPay), card and e-wallet payments
Vendor & Commercial	Vendor technical evaluation and selection, technology-investment decisions, SOW/licence negotiation, governance of IT service vendors, delivery budget
Security & Compliance	MAS TRM, secure SDLC, data-protection impact assessments (DPIA), API and database encryption, credential management, penetration testing in CI/CD, incident analysis for regulatory reporting
Methods & Tools	Agile/Scrum, hybrid Waterfall, Jira, Confluence, MS Project, Miro
Technical	Java, JavaScript (React), Spring Boot, Spring Cloud, Apache Camel, SQL, Linux (RHEL), AWS, Kubernetes, Splunk, Spark, production AI agents (Claude Code)

SELECTED ACHIEVEMENTS

- **Migration ahead of deadline:** Planned and architected the phased move of cross-border payments to ISO 20022 (SWIFT's MT-to-MX migration), running old and new message formats side by side: new-format messaging went live in May and August 2025, ahead of SWIFT's November 2025 deadline, and the old path was retired in August 2026 with no service interruption.
- **Multi-discipline leadership:** Lead two onshore/offshore platform teams through squad leads and engineering managers; earlier led VV Technology's 140-person technology organisation through its managers.
- **Vendor assessment and governance:** Lead technical evaluation of vendors, with decision or joint-decision authority on vendor selection and SOW/licence negotiation; chair governance of four IT service vendors.
- **Payment-gateway integrations:** Designed and coded the payment-gateway integrations of Renren Mall (Alipay, UnionPay) and Baidu Nuomi (Alipay, WeChat Pay, UnionPay); at VV Technology, architected a payment subsystem integrating UnionPay, Visa/Mastercard, Google Pay and GrabPay payments with settlement reconciliation.
- **Content-management system:** At Sohu, led the re-architecture of the fifth-generation news CMS and contributed to the design of Sohu News' publishing workflow — story flow, editorial review and scheduled publishing.
- **Multi-tenant SaaS:** At Golden Bridge, architected and delivered, from 0 to 1, a multi-tenant SaaS platform for online-game businesses on AWS — KYC, bonus incentives, VIP tiers and multi-channel payments — behind an API gateway.
- **AI adopted in production:** Designed and built an AI code-review service in a MAS-regulated bank, in which headless Claude Code agents review each pull request alongside SonarQube. Live since May 2026 after formal architecture and risk approval; the AI advises but never approves or merges, and senior engineers' review time fell by an estimated ~80%.

PROFESSIONAL EXPERIENCE

United Overseas Bank (UOB) | Singapore — Oct 2022 – Present

Vice President (VP2) / Technical Delivery Manager — Real-Time & Payment Data Services, Group Technology & Operations

Technology lead for cross-border payments (SWIFT ISO 20022) and retail unsecured lending, from architecture and delivery to production support, vendors and people leadership; works closely with teams in India, Thailand and Indonesia.

Technology Leadership & Delivery

- Led a 15-engineer onshore/offshore team and owned end-to-end delivery of the ISO 20022 programme — scope, roadmap, milestones, risk and issue tracking, release planning and production rollout — shipping upgrades every 2–3 months after go-live.
- Resolve schedule conflicts between the two platform lines by agreeing MVP scope with the requirement, project and technical leads, confirming dependencies and setting a sequence all parties accept.
- Own technology for retail unsecured-lending servicing — payment initiation, validation and lifecycle tracking — and post-disbursement collections across credit-card and unsecured-loan portfolios.
- Act as the primary interface between business stakeholders, product owners, operations, compliance, architects and engineering managers; present status, risks and decisions to senior management and the steering committee.

Platforms, Integration & AI

- Led end-to-end design of a payment-initiation hub that validates, enriches and tracks payments from digital and branch channels, integrates reference data and third-party banking systems, and hands payments to core banking for settlement.
- Personally built the hub's core modules in Java — payment-transaction initiation, concurrency control, the corporate- and user-role permission matrix and processing of payment status results returned by downstream systems.
- Lead the 10-engineer backend team of a regional customer-360 platform that takes daily account summaries from core banking for portfolio analysis; designed the data models and personally built the analysis module and Spring Batch jobs.
- Built an AI agent that cross-checks eMemo and quotation documents, removing a recurring manual reconciliation task; contributed to a production AI knowledge assistant used across departments, with access controls and data masking.

Governance, Security & Production

- Coordinate P1–P3 production incidents end to end — escalation, stakeholder communication, root-cause analysis and post-incident reports — and produce technical incident analyses that support the bank's regulatory reporting to MAS.
- Established delivery KPIs — deployment frequency, mean time to recovery, incident trend and change-failure rate — with CI/CD quality gates (99% SonarQube coverage gate on all code, automated regression and penetration testing) and system and user-acceptance test sign-off for every release.
- Designed security controls — API and database encryption and credential management — and led data migration for the payment, lending and customer-data platforms under secure SDLC, DPIA and MAS TRM; personally led disaster recovery (DR) and failover drills for the real-time payment middleware, sustaining 99.9%+ availability.

Golden Bridge Drive (SG) Pte. Ltd. | Singapore — Nov 2020 – Jul 2022

Project Manager & Architect

- Led a 40-person project team, including outsourced staff, introduced Agile/Scrum from scratch and CI on Jenkins and Amazon ECR, and owned delivery budget, capacity and launch readiness for the multi-tenant SaaS platform.
- Ran hybrid delivery across engineering, QA, DevOps, product and operations: Waterfall for scope, budget, milestones and resource planning; Agile sprints for execution, backlog tracking and iterative delivery.
- Architected the AWS deployment (API Gateway, EC2/ECS, RDS, DynamoDB) and applied DPIA, API and database encryption and credential management; led data migration.

VV Technology Pte. Ltd. | Singapore — Apr 2019 – Jun 2020

Head of Technical Centre

VV-Life: a local-life services platform connecting offline merchants — food, fresh produce, travel and everyday services.

- Built VV's microservices platform from scratch on Spring Cloud, Kubernetes, Redis and AWS, owning technology selection and the technology budget.
- Took part in product-strategy planning and delivered the first VV-Life release in 2019, including the payment subsystem's transaction, management and reconciliation modules.
- Led the big-data team and built VV's data-analysis platform from scratch on Flume, Hive, MapReduce and Spark.
- Designed the shared back-end platform architecture (Spring Boot, MySQL cluster, Redis, RocketMQ) for a B2B online-office product serving Southeast Asia, governing solution design, cost evaluation, staffing and acceptance.
- Introduced CI/CD, code standards and containerised deployment with rollback on Kubernetes; defined production-incident priority levels and tiered emergency-response plans.

AI Park | China **Apr 2016 – Sep 2017**

VP of Technical Department

- Led architecture, delivery, budget and the technical team for a smart-parking IoT platform connecting car parks to the cloud for automated management, online payment and recommendations.
- Led technical bids that won city parking projects in Shijiazhuang, Handan and Beijing, and launched pilots in Shijiazhuang (2016), Handan (2017) and Beijing Chaoyang (2017).
- Introduced a SaaS model for business and government customers and added an AI processing step to licence-plate and portrait recognition to improve recognition speed and accuracy.

EARLIER EXPERIENCE

- **Hengshunyuan Information Technology — Co-founder (Oct 2017 – Sep 2019, China):** Owned product, architecture and delivery of an audio-learning platform that launched with ~1,500 licensed titles and peaked at ~100k daily active users.
- **Baidu — Technical Specialist, Nuomi group-buying e-commerce platform (May 2014 – Dec 2014, China):** Led the web front-end, CMS and QA teams as project leader and delivered the order system's re-architecture two months ahead of plan.
- **Sohu.com — Technical Leader / Project Leader, Sohu News content platform (Mar 2012 – May 2014, China):** Delivered the CMS OpenAPI messaging platform and worked on Sohu website SEO.
- **Renren Network — Senior R&D Engineer (Jul 2008 – Mar 2012, China):** Developed Renren's shopping website, Renren Mall and CRM.

EDUCATION, CERTIFICATIONS & LANGUAGES

- Master's degree in Computer Science and Technology — Liaoning University (2005 – 2008)
- Bachelor's degree in Applied Mathematics — Shenyang University of Technology (2000 – 2004)
- Software Designer (Intermediate) — Qualification Certificate of Computer and Software Technology Proficiency (2006) · PMP — in progress · Site Reliability Engineering (SRE) — formal training
- Languages: English and Mandarin (professional working proficiency in both)